

Q. 1

(a)

$$\Phi_i(t) = Nt^3$$

$$\Phi_i(s) = \frac{6N}{s^4}, \quad H(s) = \frac{s^2 + s + N}{s^2}$$

$$\text{Phase error } \Phi_e(s) = \frac{s}{s + AKH(s)} \Phi_i(s)$$

steady state error

$$\lim_{t \rightarrow \infty} \Phi_e(t) = \lim_{s \rightarrow 0} s \Phi_e(s)$$

$$= \lim_{s \rightarrow 0} \frac{s^2}{s + AKH(s)} \frac{6N}{s^4}$$

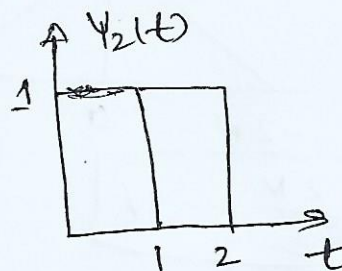
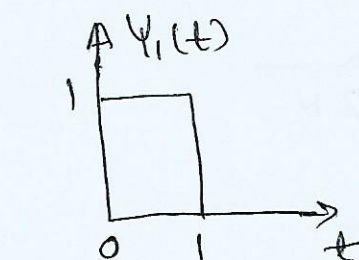
$$= \lim_{s \rightarrow 0} \frac{6N}{s^3 + s^2 AK \left(\frac{s^2 + s + N}{s^2} \right)}$$

$$= \lim_{s \rightarrow 0} \frac{6N}{s^3 + AK(s^2 + s + N)}$$

$$= \frac{6N}{AKN} = \frac{6}{AK}$$

Q. 1

(b)



3 basi



$$s_1(t) = N\Psi_1(t) + N\Psi_2(t) + N\Psi_3(t), \quad s_2(t) = N\Psi_1(t)$$

$$s_3(t) = -N\Psi_2(t) - N\Psi_3(t), \quad s_4(t) = N\Psi_1(t) + N\Psi_2(t).$$

Q. 2

$$(a) \quad S_y(f) = \frac{|P(f)|^2}{T_b} \left[R_0 + 2 \sum_{n=1}^{\infty} R_n \cos n 2\pi f T_b \right]$$

$$S_y(f) = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} R_n e^{-j n 2\pi f T_b}$$

Where

$$R_0 = \lim_{M \rightarrow \infty} \frac{1}{M} \left[\frac{M}{2} (N)^2 + \frac{M}{2} (0)^2 \right]$$

$$R_0 = \frac{N^2}{2}$$

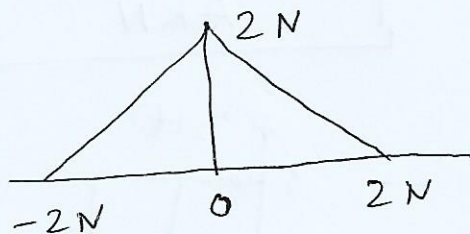
$$R_n = \lim_{M \rightarrow \infty} \frac{1}{M} \left[\frac{M}{4} N^2 + \frac{3M}{4} 0^2 \right]$$

$$R_n = \frac{N^2}{4}, \quad n \geq 1$$

and $P(f) = \frac{1}{2N} \Pi\left(\frac{f}{2N}\right)$

Q. 2 (b) $x(t) = 4N^2 \text{sinc}^2$

$$X(f) = \Pi\left(\frac{f}{2N}\right) * \Pi\left(\frac{f}{2N}\right)$$



$$R_s = \frac{2N}{2} = N, \quad T_s = \frac{1}{R_s} = \frac{1}{N}$$

Q.3

$$x(t) = P \cos 600\pi t + Q \cos(560\pi t) + Q \cos(640\pi t)$$

$$= (P + 2Q \cos 40\pi t) \cos 600\pi t$$

$$\text{Carrier Power} = \frac{P^2}{2} = 10 \text{ W}$$

$$P = \sqrt{20 \text{ W}}$$

$$\text{Side band Power} = \frac{Q^2}{2} + \frac{Q^2}{2} = Q^2$$

~~$\frac{P^2}{2} + Q^2$~~ $\frac{Q^2}{\frac{P^2}{2} + Q^2} = \frac{5 \text{ W}}{100}$

Substituting P , we will get Q .

(a) $A = P$, $m(t) = 2Q \cos 40\pi t$

$$\mu = \frac{2Q}{P}$$

(b) $x_{\text{FM}}(t) = N \cos(10000\pi t + 2\pi K_f \int_{-\infty}^t (P + 2Q \cos 40\pi t) dt)$

$$\Delta f = K_f [P + 2Q]$$

$$BW_{\text{FM}} = 2[\Delta f + f_m], \text{ where } f_m = 20 \text{ Hz}$$

(c) $\beta = \frac{2\Delta f}{f_m}$, (d) $x_{\text{PM}}(t) = N \cos(10000\pi t + K_f (P + 2Q \cos 40\pi t))$
 $\Delta f = 40Q K_f$, $BW_{\text{PM}} = 2[40Q K_f + f_m]$

(e) $SNR = \frac{\frac{N^2}{2}}{\frac{N^2}{2}} = \frac{N}{2}$

Solution

- (a) Use $\mathbf{c} = \mathbf{dG}$ to generate 16 codes.

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{Data}(\mathbf{d}) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{TransmittedCodeword}(\mathbf{C}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

For 16 correct generated codes [4 Mark] (1 Mark for each 4 correct codes). Hamming distance $d_{\min} = 3$; Using $t + 1 = d_{\min}$, we get $t = 2$ [1 Mark].

- (b) Prepare a syndrome table $\mathbf{s} = \mathbf{eQ}^T$, given in the matrix $\mathbf{S}$.

Now, calculate $\mathbf{rQ}^T$ and match with the Table to get the error pattern. Decode the coded using $\mathbf{c} = \mathbf{r} \oplus \mathbf{e}$. $\mathbf{d}$ will be the first 4 digits. Details are as follows:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{Q}^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{e} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Final answer: $d_1 = 1110$ [1 Mark], $d_2 = 1011$ [1 Mark], $d_3 = 0100$ [1 Mark], $d_4 = 1100$ [1 Mark], and $d_5 = 1000$ [1 Mark].

2. (a) $H(Y) = -\sum_i P(Y_i) \log_2 P(Y_i) = \sum_i P(Y_i) \log_2 1/P(Y_i) = 1.459$ [2 MarkS]
 $H(Y/X) = -\sum_i P(Y_i/X_i) \log_2 P(Y_i/X_i) = H(Y/X) = \sum_i P(Y_i/X_i) \log_2 P(X_i/Y_i) = 0.459$ [2 Marks]
 $I(Y; X) = H(Y) - H(Y/X) = 1$ bps [1 Mark] Or
 $H(Y) = -\sum_i P(X_i) \log_2 P(X_i) = \sum_i P(X_i) \log_2 1/P(X_i) = 1$ [2 Marks]
 $H(X/Y) = -\sum_i P(X_i/Y_i) \log_2 P(X_i/Y_i) = H(X/Y) = \sum_i P(X_i/Y_i) \log_2 P(Y_i/X_i) = 0$ [2 Marks]
 $I(X; Y) = H(X) - H(X/Y) = 1$ bps [1 Mark]
- (b) $p(t) = \int_{-25}^{-15} 25e^{j2\pi ft} df + \int_{-15}^0 50e^{j2\pi ft} df + \int_0^{15} 50e^{j2\pi ft} df + \int_{15}^{25} 25e^{j2\pi ft} df$ [1 Mark]. Since $P(f)$ is even function; inverse FT gives $p(t) = 2 \int_0^{15} 50 \cos 2\pi f t df + 2 \int_{15}^{25} 25 \cos 2\pi f t df = 100 \int_0^{15} \cos 2\pi f t df + 50 \int_{15}^{25} \cos 2\pi f t df = 25 \left[\frac{\sin 50\pi t}{\pi t} + \frac{\sin 30\pi t}{\pi t} \right] = 1250 \text{sinc}(50\pi t) + 750 \text{sinc}(30\pi t)$ [1 Mark]. Another solution can be $P(f) = 25\Pi(f/50) + 25\Pi(f/30)$. Inverse will give $p(t) = 1250 \text{sinc}(50\pi t) + 750 \text{sinc}(30\pi t)$. Since $p(t) \neq 0$ at $t = 0$, it is a Nyquist pulse [1 Mark].
From the frequency plot, $R_b/2 = 15$; $R_b = 30$, BW $B = 25$. Using $R_b = \frac{2}{1+\alpha} B$, we get roll-of factor $\alpha = 2/3 = 0.66$ [2 Marks].
3. (a) $P_e = P_0 P(\epsilon|0) + P_1 P(\epsilon|1)$; $P_e = P_0 Q\left(\frac{a_0 - q(T_b)}{\sigma_n}\right) + P_1 Q\left(\frac{p(T_b) - a_0}{\sigma_n}\right)$; $P_e = P_0 Q\left(\frac{a_0 - q(T_b)}{\sqrt{N_0/2}}\right) + P_1 Q\left(\frac{p(T_b) - a_0}{\sqrt{N_0/2}}\right)$ [1 Mark]; Differentiate P_e wrt a_0 , and equating zero to get $a_0 = \frac{p(T_b) + q(T_b)}{2} + \frac{1}{p(T_b) - q(T_b)} \frac{N_0}{2} \ln \frac{P_0}{P_1}$ [2 Marks]; Use $p(T_b) = 2$, $q(T_b) = -1$, $P_0 = 1/3$, $P_1 = 2/3$, $N_0/2 = 1$; Thus: $a_0 = \frac{1}{2} + \frac{1}{3} \log_e \frac{1}{2} = 0.2690$ [2 Marks].

(b) $P_e = P_0 Q\left(\frac{a_0 - q(T_b)}{\sqrt{N_0/2}}\right) + P_1 Q\left(\frac{p(T_b) - a_0}{\sqrt{N_0/2}}\right) = \frac{1}{3}Q(1.2690) + \frac{2}{3}Q(1.7310)$ [**3 Marks**].

(c)

$$\phi_1(t) = \begin{cases} \frac{1}{\sqrt{5}}, & 0 \leq t \leq 5 \\ 0, & \text{otherwise} \end{cases} \quad \phi_2(t) = \begin{cases} \frac{1}{\sqrt{5}}, & 5 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases} \quad [\mathbf{1Mark}]$$

4. From the diagram, number of samples 2500, duration 1 second, sampling rate 2500 samples per second, BW of signal 1250 Hz.

- (a) i. $A^2 = 50$ gives, $A = 10$, $f_c = 88$ KHz. $x_{FM}(t) = 10 \cos(2\pi 88 \times 10^3 t + 20\pi \int_0^t m(\tau) d\tau)$. Initial frequency deviation: $\Delta_i = 8$ Hz [**1 Mark**]. To get a frequency deviation of 64 Hz, an 8 multiplier is needed. This increases the carrier frequency to $88 \times 8 = 704$ KHz [**1 Mark**]. An LO of 616 KHz will generate the FM signal with a carrier frequency of 88KHz and frequency deviation of 64 Hz [**1 Mark**].
- ii. $BW = 2(\Delta_f + f_m) = 2(64 + 1250) = 2628$ Hz [**1 Mark**].
- iii. Input FM signal, followed by a differentiator, followed by an envelope detector, followed by a DC blocker [**1 Mark**].
- (b) i. $SQNR = 3L^2 \frac{P_m}{m_p^2}$. This gives $L = 128.1$, thus $n = 7$ (or $n = 8$) bits PCM [**1 Mark**].
- ii. $\alpha = 1$; this gives $p(t) = 25 \times 10^3 \frac{\cos \pi 25 \times 10^3 t}{1 - 4 \times (25 \times 10^3)^2 t^2} \text{sinc}(\pi 25 \times 10^3 t)$. [**2 Marks**]
- iii. Number of bits $7 \times 2500 = 17500$ (or $8 \times 2500 = 20000$ bits). Noise power = 10^{-1} ; SNR = 15. Capacity 100 bits/sec. Time: 175 seconds with $n = 7$ PCM (or 200 seconds for $n = 8$ PCM) for binary modulation [**2 Marks**].

Comprehensive Exam - Part B

Q. 1 (a) Let Last 3 digits are: - 3 5 7

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Code words (6)

data (k)	$d \oplus P$
0000	000
0001	111
0010	101
$\vdots$	$\vdots$
1111	001

(4)

$$d_{min} = 2t + 1$$

$$t = \frac{d_{min} - 1}{2}$$

Error Correcting Capability. (1)

Q. 1 (b)

$$G = \begin{bmatrix} I_{N,N} & \begin{matrix} C_{N+1} & C_{N+2} \\ \vdots & \vdots \\ \vdots & \vdots \end{matrix} \end{bmatrix} \quad (N, N+2)$$

$\underbrace{\hspace{10em}}_P$

$m = d \oplus P$
= module-2 operation on data bits (5)

e.g. $N=4$, $d=1111$ (even 1's)
module-2 $\rightarrow 0$

$N=5$ $d=11111$ (odd 1's)
module-2 $\rightarrow 1$

$$\begin{aligned} C(N=4) &= 1111 00 \\ C(N=5) &= 11111 11 \end{aligned}$$

Q2(a)

$$\begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} \frac{1}{N_1} & 1 - \frac{1}{N_1} \end{bmatrix} \\ x_2 & \begin{bmatrix} \frac{1}{N_1+2} & 1 - \frac{1}{N_1+2} \end{bmatrix} \end{matrix} \quad \text{Matrix}$$

$$P(x_1) = \frac{1}{N}, \quad P(x_2) = 1 - \frac{1}{N}$$

$$P(y_1) = P(y_1|x_1)P(x_1) + P(y_1|x_2)P(x_2)$$

$$P(y_2) = 1 - P(y_1)$$

$$(3) \quad H(X) = \sum \sum P(x_i, y_j) \log \frac{1}{P(y_j|x_i)}$$

$$P(x_i, y_j) = P(y_j|x_i)P(x_i)$$

$$(1) \quad H(Y) = \sum P(y_j) \log \frac{1}{P(y_j)}$$

$$\begin{cases} x_i = 1, 2 \\ y_j = 1, 2 \end{cases}$$

~~$$I(X;Y) = H(X) - H(X|Y) = H(X)$$~~

$$(1) \quad I = H(Y) - H(Y|X) \quad \text{Capacity}$$

Q.2(b)

$$\text{Noise Power } P_N = 10^{-2N} \times 10^{-3} \times 500 \times 2$$

$$= 10^{-2N} \text{ W/Hz} \quad (1)$$

$$P_s = 10^{-2N} \times 10^{-3} \times 100 + 10^{-N} \times 10^{-3} \times 200$$

$$= 10^{-2N-1} + 10^{-N-1} \times 2 + 10^{-3N-1} \times 2 \quad (4)$$

$$C = 500 \log_2 \left(1 + \frac{P_s}{P_n} \right)$$

Q3 (a)

$$\sigma_n^2 = \int_{-\infty}^{\infty} n^2 \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n|} dn = 1$$

Avg. Signal power = N^2

$$\boxed{\text{SNR} = N^2}$$

(2)

BER

First calculate the threshold

$$\frac{P_1 f(n/N)}{P_2 f(n/N)} = \frac{P_1 \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n-N|}}{P_2 \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n+N|}} \geq 1$$

$$\Rightarrow n \leq -\frac{1}{2\sqrt{2}} \log_e \frac{P_2}{P_1} = n_b^{\text{th}} \quad (1)$$

$$P_e = P_1 P(e/N) + P_2 P(e/N) \quad (1)$$

$$P_e = P_1 \int_{-\infty}^{n_b^{\text{th}}} \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n-N|} dn + P_2 \int_{n_b^{\text{th}}}^{\infty} \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n+N|} dn \quad (1)$$

Q3 (b)

$$\boxed{\text{SNR} = \frac{N^2}{\sigma^2}} \quad (1)$$

BER

Similar to Q3 (a)

$$n \geq \frac{\sigma_n^2}{2} \log_e \frac{P_2}{P_1} = n_b^{\text{th}} \quad (1)$$

$$P_e = P_1 P(e/N) + P_2 P(e/N) \quad (1)$$

$$= P_1 \int_{-\infty}^{n_b^{\text{th}}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(n-N)^2}{2\sigma^2}} dn + P_2 \int_{n_b^{\text{th}}}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(n+N)^2}{2\sigma^2}} dn \quad (1)$$

$$P_e = P_1 Q\left[\frac{N}{\sigma_n} - \frac{n_b^{\text{th}}}{\sigma_n}\right] + P_2 Q\left[\frac{N}{\sigma_n} + \frac{n_b^{\text{th}}}{\sigma_n}\right] \quad (1)$$

Q.4

(a) Average SNR.

$$\text{Average BER} = Q\sqrt{\frac{2E_p}{N_0}} = Q\sqrt{\rho^2}$$

$$\text{Average SNR} = \rho^2 = \frac{2E_p}{N_0} = \frac{2E_p}{4N} = \frac{E_p}{2N}$$

$E_p =$ Pulse energy of $\boxed{s(t) = \frac{N}{T_b} t}$ (2)

$$E_p = \int_0^{T_b} \frac{N^2}{T_b^2} t^2 dt = \frac{N^2}{T_b^2} \frac{T_b^3}{3} = \frac{N^2 T_b}{3}$$

$$\boxed{\text{Average SNR} = \frac{N T_b}{6}} \quad (3)$$

Q.4

(b) $s(t) = \frac{N}{T_b} t$, $h(t) = \delta(t-N) + \frac{1}{2} [\delta(t-N-T_b) + \delta(t-N+T_b)]$

Signal before MF, $y_1(t) = s(t-N) + \frac{1}{2} [s(t-N-T_b) + s(t-N+T_b)]$

$$\boxed{\text{MF, } f(t) = s(T_b - t - N) + \frac{1}{2} [s(T_b - t - N - T_b) + s(T_b - t - N + T_b)]} \quad (3)$$

$$\text{BER} \Rightarrow P_e = Q\sqrt{\frac{2E_p'}{N_0}} = Q\sqrt{\frac{E_p'}{2N}}$$

$$E_p' = \text{energy of } y_1(t) = \frac{N^2 T_b}{3} + \frac{1}{4} \frac{N^2 T_b}{3} + \frac{1}{4} \frac{N^2 T_b}{3}$$

$$\boxed{E_p' = \frac{N^2 T_b}{2}}$$

$$\boxed{P_e = Q\sqrt{\frac{N T_b}{4}}} \quad (2)$$